

ActInf GuestStream #021.1 ~ Shannon Proksch "Coordination Dynamics of Multi-Agent Interaction"

GuestStream | Shannon Proksch | 1:20:59

Hello, this is ActonFlab guest stream number 21.1. It's April 21st, 2022, and we're here with Shannon Proksch, and Shannon will be giving a talk, Coordination Dynamics of Multi-Agent Interaction. The talk will happen for maybe around an hour, and then there'll be time for Q&A, so feel free to ask any questions in the chat during. So off to you, Shannon, looking forward to this talk, and thanks again for joining on ActonFlab.

Awesome, thanks so much. So I'm Shannon, Daniel's already given you the title of our talk, and I'm coming to you from the University of California in Merced, where I'm a PhD candidate, and I'll actually be defending this summer, so maybe the next time you hear from me, I'll be talking to you from Augustana University, where I will be starting a tenure-track professor position later this fall as an assistant professor of psychology and neuroscience.

Oh, applause, nice, thanks. So that said, I'm not going to talk to you about neuroscience today. I'm going to talk to you a little bit about the behavior of crowds, and I stole this roadmap idea from previous Active Inference Lab sessions, and this is kind of outlining where we'll be going today. So first, the first thing we're going to do is really dig into a little bit of theory and data.

I'm going to give a bit of a brief introduction to multi-agent interaction and what kind of human behavior I'm interested in. And I'm going to walk through a couple of ways that active inference folks might be interested in crowd dynamics and behavior, but primarily, I'm going to be focusing on dynamical systems theory and the notion of synergies and the tools that come along with these concepts. And then finally, I'm going to show you a model system where we apply these tools to a system of coupled metronomes before I look at some empirical data from real world interaction, musical performance, and also of crowds cheering at a basketball game.

So first, what is multi-agent interaction? This is simply an interaction that's happening between two or more agents. In our case, we're looking at people and each of these people, they'll have their own personal goals and behaviors, but their individual behaviors are influenced in some way by interacting with other agents in their shared environment. And I want to talk about two forms of multi-agent coordination today. And the best way to introduce this is through a short thought experiment. So I'll just have you imagine the sounds of a crowded coffee shop. Consider how individuals in that coffee shop might be interacting with each other. There might be some small groups, there might be pairs, but mostly it's many individuals engaging in small temporary interactions. And these individuals aren't coordinating with every other individual in some sort of cohesive coffee shop group. They're just a jumble of individual individuals who are cohabiting a shared space.

Now instead, imagine the audience on the floor of a rock concert. They're cheering or singing along with the artists on the stage. And alternatively, imagine the fan section at your favorite sporting event emerging into a synchronous chant or a chorus of resounding booze. Imagine how individuals in these large crowds

might be interacting with each other. As they cheer, sing, chant, or boo, they're all sharing in similar behavioral states, engaging in similar actions. They're likely sharing similar physiological states, like breathing rates, and neural states as well. These crowds are changing together in time. They're behaving and they're coordinating like one large interdependent group. And you could refer to these as shared acoustic spaces. And they include various levels of interaction from dyads, pairs of individuals, or larger and larger groups. And I might even call these acoustic social worlds. The acoustic signal that's generated by these crowds, it carries some information about their behavioral dynamics that were employed in creating that signal. And the sounds these crowd makes carry some sort of information about the status of that crowd, either as an incidental collection of lots of individual coffee goers, or an integrated and coordinated crowd, like a crowd of concert goers. And there are, of course, many important factors at play when we're evaluating these acoustic social worlds. And we're trying to decide whether that crowd is acting as one integrated multi-agent system rather than a collection of independent agents. And these include the social context, a coffee shop is different than a concert arena. The development of patterned cultural practices in these different contexts, and differing scripts surrounding what events happen, and when these events happen, and differing regimes of shared attention between each of these agents. And these are things you've talked about in the active inference lab before. And basically, this is all a rather long-winded way of saying that we expect coffee shop behavior in a coffee shop and concert behavior in a concert arena. If I were going to talk to you in the language of active inference, I might talk to you today about Markov blankets. And we might talk about this statistical boundary that might be behind a pair of interacting individuals around a pair of interacting individuals separating their internal states of this interaction with the rest of the interactions happening in a coffee shop crowd. And I might talk about how the statistical boundary expands as we start to engage or interact with larger groups of people from three to four to maybe 100 people or more if we're in a large group interaction. And while this can be a really useful way to model multi-agent interaction, to talk about a suite of nested Markov blankets, we quickly run into questions of how a researcher chooses where to draw a Markov blanket at any given system, or where to draw each nested Markov blanket in a given system. And this is a topic that you've discussed a lot in the active inference lab. You've spent some time talking with Yella Brunenberg and colleagues after their recent BBS article commentaries and how there might be differences from me as a researcher applying some sort of Markov blanket around a system versus the actual agents interacting in that system. But I just mentioned this idea of Markov blankets to situate the topic for the audience of these interactive inference lab live streams. As far as measuring the emergence of an integrated interacting crowd, this is an empirical question that I'm asking rather than a modeling question. And I'm not a mathematician or a modeler. So for this question about groups of interacting people, I found empirical tools from dynamical systems theory and with the concept of interpersonal synergies. So that's what I'm going to introduce to you guys next. So dynamical systems theory, this considers a trajectory of behavior of a system over time. And the concept of synergies arises from the field of motor control as a solution to the degrees of freedom problem. So essentially, the degrees of freedom problem is that fine and precise control of every degree of freedom in a motor movement would be computationally intractable for

the nervous system to solve. And what you see here is a minor hammering. And you see these light patterns. This is really sort of the earliest sort of motion capture that we had. And we're seeing the trajectory of his arm hammering here. And you can see that even though the hammer is hitting the same place every time, you'll have to trust me that's hitting the same place every time, his arm is taking a slightly different trajectory to get there. But it generally covers this one swinging behavior.

And that's because the muscles in his arm here are self organizing into coupled and locally constrained degrees of freedom, forming temporary soft assemblies of motor function that's enabling this flexible movement and coordination here. Put another way, synergies are identified by their functional specificity, dimensional compression, and reciprocal compensation. What are those? Well, the need to perform a specific

function like hammering that anvil, that functional specificity temporarily shapes the coordination of several components into one system. That's dimensional compression. In which external and internal perturbations are flexibly dealt with via compensatory adjustments of the components, reciprocal compensation.

So each muscle in his arm is compensating for the activity of the other muscle. And if he was bumped slightly, there would be compensation from other muscles to ensure that anvil hits its location. And this is all done in order to preserve the functionality of the whole system for as long as it's needed. And that last comment for as long as it's needed, that gets to the notion of soft assembly, where any of these units in this picture here might be functionally coupled just for as long as you need to perform that task. So if we're talking about a group of people, this group of people might form some sort of complex system just while they're interacting for that meeting, while they're at that concert, until they disperse and are ready to move on. We can differentiate mere coordination, which might happen between two people as the result of engaging with the same stimulus and similar patterns of actions from a true interpersonal synergy. And you see here, we

might see the same patterns of movement if we have two individuals who are maybe just listening to the same set of

music. But if these individuals are interacting with each other, where the action of one person and the action of

the second person is constraining each other, then this is where an interpersonal synergy happens. And these synergies are the result of these micro scale structures of a system,

giving rise to macro scale dynamics. When a synergy develops, we no longer need to describe the independent dynamics of each micro scale structure. We can instead describe the behavior of this single interaction dominant system,

which consists of complementary actions from each person's motor activity as it pertains to their role in this softly assembled complex system. And you might even notice some similarities here between these graphics and a graphic we might draw of nested Markov blankets. And these might very well be two sides of the same coin, just two perspectives with a very similar conceptual landscape here.

And there's a variety of methods for studying the coordination dynamics that underlie interpersonal coordination. For example, we might ask two people to sit next to each other and swing pendulums or two rock and rocking chairs.

And we can measure how these individuals intentionally or unintentionally synchronize their behavior.

As each individual's action becomes coupled to their partner. Or we can ask musicians to play together in a musical ensemble or a duet.

And we can measure how these quartets or duets move together in time, forming these loosely coupled interpersonal synergies.

And in each of these interactions, a motor or acoustic behavioral output of each interacting individual was measured and analyzed for meaningful correlations between these individuals.

So importantly, in these experiments here, it has been possible to obtain clear measurements of individual behavior.

In order to correlate the behavior of these individuals and examine the emergent coordination dynamics of multi-agent interaction and social self-organization.

But what happens when we have this massive crowd of people?

Can we measure the interpersonal synergies that form in such a large crowd?

How does a group of people move between novel and habitual patterns of coordination?

How does a multi-agent system navigate transitions between chaos and stability?

And let's make it a little bit trickier.

The real world is messy.

The measurement is difficult.

In a naturalistic social interaction and a very large crowd.

I probably can't measure the behavior of every single individual person.

I probably can't give everyone an individual microphone.

I probably can't track each person's individual brain states.

And I might not even have access to a camera, let alone a set of motion capture cameras and hundreds of markers to track everyone's head and limb movements.

So let's come back to the crowds we imagined at the start of this discussion.

When you're imagining this coffee shop crowd and the crowd at the rock concert, you might have also been imagining how these concerts, how these crowds sounded.

So the difference between this disjointed noise you might hear at the coffee shop and this altogether coordinated singing maybe at a rock concert, it's trivial, trivially easy for you or I to identify.

An uncoordinated group of independent individuals happening to coexist in a shared space versus a coordinated group of interdependent group members in a crowd.

We can simply hear that these two groups of people sound different.

Similarly, we can simply hear when we ourselves are engaged and participating as a collective sort of agent while we're interacting with a large group of people or in a very large musical ensemble.

So I'm taking that idea.

And I think we can measure the sounds that a crowd makes and with a well placed microphone or a few microphones, we can generate a single audio recording of the sounds that an interacting group makes over time.

And I know that a human can make inferences about the interactions and the acoustic social world of a crowd just by listening.

So similarly, I want to know if this dynamical systems toolkit that I have can allow us to make inferences about the behavior of an entire complex system based on this one acoustic measurement that we have access to.

So for the next little while, I'm going to walk through a bit of theory and then a toy example until I eventually end this talk with data from real live multi-agent groups that we're analyzing with tools from dynamical systems theory.

So one of our most simple complex dynamical systems is a metronome or a pendulum.

So Christopher Huygens first discovered this activity of two pendulums from pendulum clocks will be related to each other when these clocks are hanging from a shared surface.

And what he found was that these pendulums would swing in anti-synchrony with each other.

And since then, in 1600s or so, there's been lots of research on pendulums and on coupled metronomes.

And if we put these metronomes, a timekeeping device for making music on a shared platform and maybe that platform rotates like this,

you'll see each of these metronomes come into synchrony with each other.

These are just static pictures. So to really help you visualize this and hear this, I'm recruiting a little bit of help from the Mythbusters.

The Mythbusters on YouTube, so hopefully it is free and uncopyrighted.

And I'm just going to skip ahead here to partway into their episode.

Friction and momentum as I can so that only the metronome is having the effect.

Now let's try two.

Adam doubles his trouble with two metronomes set to the same tick mark.

Look at the movement on this platform.

And after two minutes, the shimmy takes over.

Stay with it, boys.

Nice.

Nice.

Nice.

It's totally working.

It just took them a little while to find each other's rhythm.

Cool.

I have an idea.

If I stop the platform, they should drop out of phase.

There it is.

The moving platform is totally, totally critical to this experiment.

So.

All right.

So that's Adam Savage there with two metronomes.

One on earth.

Why does this matter?

Well, what he's done is replicated the findings of that centuries old experiment that was originally conducted with pendulums on a clock.

And he noted that this moving platform here is what's totally, totally critical.

But what I want you to also think of is that you heard these metronomes slowly coming into phase with each other.

They were becoming interdependent on each other's behavior because they are connected by this coupling mechanism here.

This the shared platform.

And as Adam correctly notes, this moving platform is totally, totally critical.

And in fact, when you saw him stop the platform, the metronomes are no longer coupled and their behavior no longer affects the other.

This platform here, it can take many forms as long as it is freely flowing and responsive to the metronomes movement.

And for the metronomes, this platform is the essential coupling mechanism for this emergence of coordinated acoustic behavior.

In humans, this coordination doesn't have to be a physical platform.

In fact, I argue that it is the interaction itself which serves as a coupling mechanism for multi-agent human behavior.

And we're going to spend just a little bit more time with metronomes before I get on to real human groups here.

So when you have many metronomes on a shared platform, you can have more than two.

They're going to transition from uncoordinated random phase to coordinated synchronous phase.

And what you're going to hear next is the acoustic output of 32 metronomes on a freely moving platform.

And I'm going to justify to you two things that we can identify emergent coordinated behaviors in acoustic data.

And what all I'm going to justify what state space reconstruction is and how this tool from dynamical systems can help us to analyze that acoustic data.

So I'm going to start at the beginning with each of the metronomes in random phase.

And this really takes a bit of a time so we don't have to listen to that for so long.

I'm going to skip ahead.

So what you heard was 32 metronomes all ticking at their own tempo at their natural period.

It sounded like a bunch of random noise.

But this platform, again, served as a coupling mechanism and synchronization between the metronomes emerge.

So they began to sound like one group ticking together.

And this is an example of the type of data we'll be investigating for the emergence of coordination and acoustic signals.

So keep these metronomes in mind.

No, don't play the metronomes again.

The platform is serving as a coupling mechanism.

As synchronization emerges, they begin to sound like one group.

So we're going to visit some theory.

I'm going to introduce you to Totkin's theorem and phase space reconstruction.

So if we take one time series from a system, this is the Lorenz system here.

It's defined by a set of three differential equations.

But what you need to know is that we can reconstruct the behavior of this system as a whole by measuring only one of its parts.

This is accomplished by embedding delayed copies of a single time series into multiple dimensions on some delay.

So here's one of these time series extracted from the Lorenz system.

And we delay that time series to make another time series, delay once more to make a third time series.

We take these three dimensions and we use them to reconstruct a projection of the Lorenz system.

And this is phase space reconstruction.

And you can see that it maintains the topography or the same shape as the original Lorenz system.

In the case of our acoustic data, I'm going to take a time series of that sound, that acoustic signal.

And I'm going to follow these steps here to reconstruct the behavior of those metronomes over time.

And once we've completed this phase space reconstruction, I'm going to visualize the behavior of our reconstructed system by generating what we call recurrence plots.

This is done by plotting recurrent points from the reconstructive phase space within some radius that we set.

And we can evaluate the structures in these recurrence plots using recurrence quantification analysis.

So on the right here, you're seeing a few examples of different time series.

And you're seeing the reconstructed phase space here.

And on the bottom, you're seeing the recurrence plot that we generate from these reconstructed phase space.

So one of the simplest measures that we can glean from recurrence plot is the rate that states recur.

And this is called recurrence rate.

And on the left here, we're seeing a time series of a simple sine wave.

And when we reconstruct it, we end up seeing something a bit like an oval or a circle.

And when we plot a point on our recurrence system, every time our recurrence plot here, every time our system revisits a space on the circle here.

So in the sine waves recurrence plot, we're seeing these diagonal lines.

These diagonal lines are reflecting deterministic behavior.

And they track how similar sequences of values change over time.

So this sine wave here, it's reliably deterministic.

If it starts here, it's going to go here next.

And it's going to go there next.

And that gives us these long recurrence values here.

And as you probably expect, this measure is called determinism.

In the middle, we see uncorrelated white noise.

And you can see the signal already looks much more random than our sine wave.

And when we reconstruct it in phase space, I don't really see a shape here that we can get much out of.

And underneath, you see our recurrence plot.

And it looks similarly random.

You could think of it like white noise on your television even.

Finally, on the right, we see a time series extracted from speech.

It's a little bit deterministic, like our sine wave.

If it goes up, it's going to come down.

But it's also drifting a bit.

And you can see if we reconstruct that phase space instead of a simple oval, we're seeing a sort of transient signal here.

And in the recurrence spot, we're also going to be seeing variation in lengths in these diagonals.

And finally, recurrence spots can contain these vertical structures that you don't see here in addition to diagonals.

And these are indications of periods of stability.

We measure these with a value called laminarity.

And laminarity, you can contrast it with determinism.

It measures how a signal is organized in terms of simple, similar absolute values that are stable over time.

And this is one representation of recurrence spots.

It takes a little bit of time to get comfortable with what a current spot is.

So I just have a different visualization to show you a little bit of the same thing.

So these are the same set of RQA measures that we take from our system and plot on recurrence spots.

This is demonstrating recurrence rate.

It's just the percentage of recurrent points that we see on this recurrence plot over time.

And it represents patterns of behavior that repeat over time.

Determinism is the percentage of points that fall in a diagonal at any time.

And it represents behaviors that belongs to a longer sequence of behavior.

Entropy, this is the variability in lengths of these diagonal lines down here.

And it represents the amount of disorder that there might be in these sequences.

Finally, laminarity here are the percentage of points that fall on a vertical line.

And this represents clusters of behavior that might repeat for a length of time.

Like when a system revisits a period of behavior, leaves and returns to that behavioral state again.

Alright, so it's a lot of theory.

I need to bring you back to the metronome so we know what's going on here.

So what you see here, this is a picture of our uncoordinated metronomes.

They're moving in all sorts of directions.

And finally, our coordinated metronomes at the bottom.

And I took a 30-second acoustic signal from each set of behaviors here.

And on the right, I have signals from this uncoordinated metronome and the recurrence spot we generated from them.

And you can see this first recurrence spot, it resembles that kind of uncorrelated white noise.

I don't see a lot of structure here.

And then in our time series for the coordinated metronomes, you start to see some periodicities kind of in this signal here.

And you see a lot more of these laminar structures down here in the bottom, these vertical structures, which indicate these periods of stable states in this group metronome behavior.

I could break that down even smaller and I could look at five seconds of sound.

And we see the same structures emerging.

You can also see a bit more clearly these spikes that occur in the coordinated metronomes that correlate with these vertical structures than we did in the last plot.

So this is great.

Now you're experts.

You've learned everything you need to know to evaluate the emergence of coordination when we measure it from a global acoustic signal.

And you know how to visualize it in a recurrence plot and how to interpret that plot.

So we're going to apply all of this to human data.

And I want to remember for the metronomes, this platform is the essential coupling mechanism for the emergence of coordinated behavior.

For humans, the interaction is totally, totally critical.

So say me and my colleague, Marlee Reeves from the Applied Mathematics Department here at UC Merced.

So we don't need a shared platform.

Instead, we just need to interact with other human beings in order to couple our multi-agent human behavior.

So now that we've got this far, I promised you some empirical data.

And I'm going to tell you about multi-agent interaction first in a specific musical ensemble performing a specific musical work.

And then in the crowd at a basketball game.

So first, our musical ensemble here.

They're performing a work called Welcome to the Imagination World by Tasuki Shimizu.

And this group of musician, they're performing this piece that was composed to do something interesting.

So the composer wrote the first few minutes of this musical piece to be completely random.

In music theory terms, we call this alia torque.

It's when some element of the music is left up to chance or to whatever the performers decide to play in that moment.

So the start of this piece sounds a little bit like this.

So.

So this is not the musicians warming up on stage.

This is the actual music.

So at the top, you're seeing the first and last 30 seconds of that first piece, that first part of the performance.

On the bottom, you're seeing the first and last 30 seconds of the second part of the performance once the musicians have become coordinated.

And we can see that the acoustic behavior from these musicians in the coordinated section of the music is more structured.

We see these bursty vertical lines here where you have these stretches of connected sound, connected stretches of sound for periods of time.

And again, we can look even closer at five-second samples, and we can see the same structures in, same vertical structures in the coordinated, but not the uncoordinated sections.

So if I take just these little blocks here, zoom in to five seconds, we see a little bit more structure here, and a little bit less or a little bit noisier structure here.

These are just for visualizations.

This is not actually the five-second snippet here, but it works for getting the idea across.

Okay, so cool.

I see these structures in the recurrence plot, but is it really more structured when musicians are coordinated?

We can do some statistics, and we can visualize our data in another way.

So these are distributions of all of those measures that we extracted from the recurrence plot.

And you can see coordinated sections in blue, uncoordinated sections in red.

And we're seeing that when the musicians begin interacting with each other as a coordinated musical ensemble, recurrence and stability measures increased overall.

So we have higher, also more variable RQA measures when they're coordinated, and we see lower and less variable RQA measures when they're uncoordinated.

And we can see this even a little more clearly if we can plot those same measures over time.

And the blue and purple lines here are representing uncoordinated section of the piece.

Orange and green is represented.

Sorry, blue and purple is coordinated.

Orange and green is uncoordinated.

And we can see that over time,
when the musicians are coordinated
and acting like one interacting ensemble,
they're able to explore a greater variety
of recurrence dynamics.

They're able to explore more behavioral states
than when they're uncoordinated.

And we see all of these measures,
no matter what they are,
hovering around a small subset of dynamics.

Now, you would be forgiven
if you are skeptical of these results.

After all, the entire musical performance
is over eight minutes long.

Maybe I simply chose the most uncoordinated sections
and the most uncoordinated sections of the song.

And you'd be smart to have this worry.

Our peer reviewers shared the same worry.

So we graphed the time series of RQA metrics
for the entire piece.

And it's pretty clear,
even if you don't know what these values are,
that there's some difference
in all of these top pink and blue kind of graphs
and these bottom orange and green type of graphs.

We're seeing this same greater variability,
greater values and more variability
in each of these RQA metrics
when the musicians are coordinated
than when they're performing
as a bunch of individuals on stage.

And the takeaway here
is that these musicians,
when they're interacting together
as a musical ensemble,

they're able to explore
a greater variety of acoustic behavior
when they formed a single complex system,
an interpersonal synergy,
then when they were acting
as a lot of individuals on stage.

And we were able to use
these nonlinear statistical methods
from dynamical systems theory
to measure that change.

Cool.

So that's a musical ensemble.
I'm using it as a model system
because we know what the behavior should be.

Can I measure this
from a different kind of system,
one that doesn't have a script
dictating the behavior
of what's going on on the stage?

So to do that,

I look at data recorded
from the student section
of a BYU basketball game.

And this data was collected
by Butler and colleagues
and labeled by many hardworking
undergraduate students.

And we have basically recordings
from the crowd
that are labeled
based on what kind of crowd sounds
they're making,

whether they're cheering,
this is like a loud,
positive crowd vocalization,
whether they're applauding.

So this is mostly clapping,
but could include some vocalization.

Distraction noise

when you're trying to distract someone
from a free throw,
mainly positive chants.

So this is the fans chanting
their BYU cougars chant
or defense.

Negative chanting,
it's when they're angry
rather than cheering positively,
singing and then silence.

So we're taking this data
and we're going to ask
a similar question.

So we don't have a clear transition
from uncoordinated
to coordinated behavior,
but we do have all of these
different types of behavior.

So we're going to ask,
is there a relationship
between nonlinear recurrence dynamics
measured by RQA
and these different categories
of acoustic behavior
at the crowd
at this basketball game?

And we follow the same process.

We extract acoustic signals
and we cut them up
into five second samples here.

And we use these signals
to generate recurrence plots.

And we take the measures
from that recurrence plot
to see if there's anything different.

And I'm going to show you
just two recurrence plots

on the next slide.

I'm going to show you a plot

from positive chanting

and from distraction noise.

So here's distraction noise.

It looks like white noise.

It looks like what you would see

on your television screen.

And when I show you positive chant,

there's clearly a striking difference.

There's almost no structure

in this distraction noise here,

but there are very clear

laminar bursts of activity

represented in these vertical lines.

And in fact,

these two plots look very similar

to our unsynchronized

and synchronized metronomes

way back from the middle

of this talk.

And intuitively,

this makes sense.

If you're trying to distract someone

during a free throw,

you're making all sorts

of random sounds.

You might be shaking

your keys around.

People are stomping on the stands.

But if you're cheering your team on,

trying to motivate them

to score more points,

or if you're mounting

a strong defense,

you're going to be working together,

shouting whatever your chant is.

Go, team, go, defense.

And it's going to occur
in a very rhythmic metronomic fashion.

And we can compare
how all of these distributions
of our QA metrics look
between each different
kind of crowd noise.

And you can see
there's quite a lot of overlap,
but there's also some difference.

So positive and negative chant
seem to be a bit similar
to each other in every category,
which makes sense
because whether they're chanting
in a positive or negative energy,
it's still this rhythmic,
isochronous behavior
that's happening.

Distraction noise,
cheer, and applause,
on the other hand,
all seem pretty similar
to each other,
no matter where we look at them.

And then finally,
all the way on the lowest scale
is angry noise here.

And one thing to note
is we had very few samples
of angry noise here.

So that could be why
we're seeing a few
of these differences.

And if we had more samples,
maybe angry noise
would look a bit more
like distraction noise.

So visually,
we can see there are differences
in these RQA metrics
for all of these crowd sounds.
And we don't want to just do visuals,
so we can do some statistics here.
This is just another visualization
of our estimated marginal means
showing pairwise comparisons
between every crowd sound category.
And it's a graphical representation
of the estimated marginal means.
If these red lines overlap,
that means there are not
significant differences
between, for instance,
distraction noise,
cheer, and applause here.
Positive and negative chant
do look different, however.
Interesting.
And then angry noise
is far out by itself.
So knowing this,
I mentioned very briefly earlier,
I have a colleague
in the Applied Mathematics Department.
And one thing she's interested in
is whether we can give
a machine learning algorithm
some of these RQA metrics
to tell the difference
between these different crowd sounds.
The researchers at BYU
have done this
with traditional sound metrics,
spectral analysis,
captural analysis,

and we're doing it
with our RQA metrics.
So we're focused more
on the rhythmic content
rather than pitch
or spectral content
of the sound.
And she is doing this
in two ways.
So one is training
a neural network
with image classification.
And what I'm going to show you
a few results from
is a random forest
that is using
these recurrent features
as input
and doing some feature
vector classification
using decision trees
and doing some stuff
that if you ask me
questions about
I will not know the answer
because I'm not the mathematician
in this context.
But once she's trained
that algorithm,
it can actually learn
the differences
between these crowd sounds.
So if these are dark blue
along the diagonal,
that means that it
categorized cheer very well.
If they're light blue,
that means there was

some confusion here.

And if we remember
through one of the
previous graphs,
applause,
distraction,
noise all kind of
look similar to each other.

So if we sort of
smush those
into one distraction category,
put applause and cheer together
and put positive
and negative chance here,
we get even better classification.

And this part is something
we're still working on.
Hopefully we will be ready
to submit it in a paper
kind of soon.

But now that we're ending
on this really complex point,
what have we learned?

First, we took a
quick dive into
dynamical systems theory
and the notion
of interpersonal synergies.

I told you that
interpersonal synergies form
when a functional coupling
emerges between the behavior
of two or more individuals.

This kind of synergy
can arise
when individuals
begin to coordinate
and as an integrated

multi-aging group of people.

But measuring
this emergence
of coordination
in large groups
of people
is difficult.

Even if we can
intuitively hear
the difference
between a coffee shop crowd
and a crowd
at a concert
or sporting event.

Without individual signals
from each person
in this interacting group,
it's very hard
for scientists
to measure
when the coordination occurs.

But if we can hear
this difference,
then maybe
we can measure it.

So we practiced
with a toy model,
a pair of coupled metronomes,
transitioning from
uncoordinated
to coordinated activity,
unsynchronized
to synchronized.

And then we applied
these same
nonlinear statistical
analysis techniques
from dynamical systems

to measure
uncoordinated
and coordinated behavior
in a musical ensemble.
Transitioned once too far.
Ah, my musical ensemble
is behind our graphs here.
Apologies about that.
I can come back
to this slide.
So when we want
to learn something
about social behavior,
music provides
a good model system.
You know roughly
and can manipulate
what roles
each member
of the musical group
is performing.
And sometimes you even
have the aid
of a musical script
like we did
to be able to carefully
measure the coordination
dynamics that emerge
in this joint music making.
These same
intentional interdependencies
which arise
from a group of musicians
co-creating music together
can also emerge
as the result
of local interaction
between individuals

in large interaction
groups of people
like crowds
at a basketball game.
And we investigated
that by measuring
the different crowd sounds
that these fans made
and the different
acoustic behavioral dynamics
we could observe
in each of these crowd sounds
with fans
at a BYU basketball game.
And all this work
was not done just by me
but also my colleague
Marley Reeves
in the applied mathematics
department,
my advisor
Mesh Bell-Supermanium
and also our colleagues
Chris Kello
and Michael Spivey.
So that's all I got.
Thank you so much
for having me.
Awesome.
Thank you.
You can unshare
and I'll ask some questions
and if anyone
in the live chat
has questions
they can also ask.
Sounds great.
Are you going back

to the slide?

I was going to go back

to the slide.

I also cannot find

the unshare button.

Ah, it's fine.

It's in,

it's in,

it's the same

as the share button

but yeah.

Cool.

Oh.

All right.

Well,

a lot of

very interesting research

and really appreciate

the care

and to kind of

speaking it

very concisely

and clearly.

So just if I could

give one

takeaway

that I thought

was really fascinating.

So you,

in the Fusiroli

2014 quotation,

you talked about

how there was

like synergy

and about

the continuum

and the spaces

that are

uncoordinated
versus increasingly
coordinated
from the sheet music
which is one type
to the emergent
coordination
and then
I thought
that was
an interesting way
to open up
coordination
not in a framework
of either
strong versus
weak emergence
but rather
within a
directly measurable
dynamical systems
perspective
and then
at the end
to do the
machine learning
and be able
to train a model
on the recurrence
statistics
and then
identify the system
it's like
there's information
about the system
state
as identified
by experts

which is what
that labeling
task was about
that's retained
like just
in the coordination
pattern
and so
there's something
about the patterns
across systems
that can be
generalized
that captures
also important
information
and that's an
intuition that I
think people often
bring up
when talking
about dynamical
systems
but I really
thought it was
made clear
with how you
presented it
yeah thanks
I'm glad that
that got across
that was kind
of a message
I was trying
to share
one place
to maybe start
is this

soft assembly
so where
does this
soft assembly
come into play
and I guess
just what
does it mean
in human
teams or
communication
yeah
so this
this notion
of soft
assembly
let me think
of something
that's not
soft
something
that's
hardwired
oh I don't
have I don't
have an opposite
example to soft
assembly
so if you
take if you
take like
a tool
like I don't
know a wrench
it only can
move in one
way the parts
of the wrench

or the parts
of a set
of pliers
they're not
softly
assembled
they're always
attached
and they're
always going
to perform
that function
right
but if we
think about
lots of kind
of systems
if we think
about human
systems
if we think
about ant
colonies
or if we
think about
anything where
an individual
member of that
system could
leave if it
wanted to
or could
leave when
they're not
interacting
that's what
we would
call a

softly
assembled
system
and a
group
of humans
the example
that I was
thinking of
was any
group of
humans that
comes together
for some
amount of
time if we're
talking about
musicians on
stage they're
together and
they function
as a string
quartet or as
an orchestra
so long as
they're on stage
and performing
together but
then they can
pack their
instruments up
and they can
go home
so there's
not a strong
bond or like
a strong
physical

mechanical
connection
that's forcing
their interaction
to occur
it's just
happening
during that
process
because you
also connected
that to like
the transient
motor chains
and so that
was pretty
interesting
again anyone
who's watching
live can ask
a question
otherwise I'll
just keep on
asking about a
few points
so you use
the metronome
as a
example of
synchronization
but also
there's listening
to repetitive
music
whether it's
like electronic
music that's
repetitive or

other kinds
of music
binaural beats
so what
is this
repetition
listening
nature
and what
like kind
of makes
us select
a given
level of
repetition
to desire
like what
how do
we think
about that
in terms
of how
you presented
it with
the recurrence
here
yeah so
there's two
things that
I would
I would
think about
so first
you mentioned
binaural beats
and we
want to listen
to these

very repetitive
musical patterns
and so there's
two ways
that our brains
might be
interpreting
rhythmic patterns
in music
one notion
that is probably
very amenable
to the active
inference lab
and what you
talk about
is predictive
coding
and the idea
that we have
these hierarchical
predictions that
are happening
in the brain
so when we
perceive this
low-level stimulus
this rhythmic
input
we're constantly
predicting when
the next input
is going to
arrive
and we're only
passing up areas
when our prediction
is wrong

and isochronous
these very rhythmic
beats
are really easy
to predict
but there's
another option
it could be
that we're not
doing any prediction
at all
and we just
have these
inherent
oscillations
these natural
frequencies of
oscillations
in our brain
and when we
hear a rhythmic
pattern
that is similar
enough
we naturally
entrain
to that beat
or that rhythm
drives this
oscillatory behavior
and so you have
this emergent
entrainment
and no predictions
necessary
and you could be
in either camp
I alternate

between being
a firmly
predictive coding
camp
and being a
entrainment
camp
but this
either way
what seems
to be really
important in this
activity
is activity
of your motor
system
and your ability
for your motor
system
which is really
good at making
periodic movements
like moving
walking
the ability
of the motor
system to help
the auditory
system
to communicate
with the auditory
system
in predicting
when the next
beat occurs
or in
entraining
to the beat

of a sound
like that
and that's
with binaural beats
the second thing
that I would
bring up
is we like
these binaural beats
maybe they're
relaxing for some
people
but we're
listening to music
there's a phenomenon
called musical groove
and this
is when
it's very
scientific
no it is
actually the
scientific term
Peter Janata
in 2012
coined this
and it's
when the rhythm
of a music
is just
complicated enough
that it makes
us want to
move
to music
so it's just
in that sweet
spot where

it's predictable
but not quite
predictable enough
and it engages
our motor system
instead of just
passively engaging
it
it actually makes
us move
to the beat
of the music
this might be
because when we
move
when we tap
our foot
when we dance
we're better
able to predict
when the next
beat happens
thanks to
the groove
again
jargon
so apologies
it
it
reminds me
of the zone
of proximal
development
and kind of
what does it
mean
if there was
a build-up

and too much
tension being
built
like a ramping
but then there's
for example
when there's
quarter notes
for two measures
and then eighths
and then sixteenths
and then it's like
when it goes
even faster
and then it
it's like a meme
with making it
more exciting
because it's
reflecting a
cultural
and potentially
related to
biology
like our priors
on increasing
rate of
the
the sound
is like meaning
something
because it can't
go on forever
so then something
is exciting
or something
is changing
so then that's

very interesting
about how our
like what is
quiet
what is loud
what do those
represent
and what are
the priors
of the musical
experience
that becoming
cultured in
different ways
so
yeah
maybe how does
it help us
think about
like different
preferences
amongst individuals
in the kind
of music
that they want
to play
or engage
in or listen
to or how
they want
to engage
with that
music
yeah
so
so much
so much
content

in what you
just said
um
the one
is
is this
like notion
of anticipation
when the rhythm
keeps getting
faster and
faster
um
and
there's
um
a whole
book by
leonard meyer
about
um
how
music
is correlated
with emotions
and how
there's this idea
that a composer
is playing
with these periods
of tension
and release
so whether
that tension
is from
um
buildup
of rhythmic

events
and we
usually
expect
um
the faster
rhythm gets
maybe
it's associated
with
a certain
activity in the
music
or we expect
it to slow
down after
it's gotten
faster for so
long
or you're
thinking about
um
when the beat
drops
in a piece
of electronic
music
for instance
um
we can play
with the rhythm
or we can play
with the musical
pitch
and we might
have a certain
set of chords
that usually

lead up
to a certain
resolution
and we can
trick everyone
in the audience
if we play
a different
chord instead
um
but you mentioned
this kind of idea
of different
individuals
liking different
kinds of music
and this
zone of proximal
development too
so this is
really getting
into how
the music
that we're
exposed to
shapes the
expectations
that we're
going to have
um
so if you
grew up
hearing western
music
if you
are trained
in western
classical music

you have
certain
expectations
of what
rhythms
are allowed
of um
what kind
of melodic
or harmonic
progression
is going to
happen
but when
you hear
a different
kind of music
it might
not even
register
to you
as music
the first
time you
hear it
until you
learn about
it
until you
learn about
the music
of that
culture
um
even if
it's
music
within

your own
culture
say you
listen to
uh
hard rock
for the
first time
or like
like heavy
metal for
the first
time
it might
just sound
like noise
until you're
exposed to it
enough
or if you
study the
patterns of
of how
that music
um
moves in
time
it's like
the
plastic
learning
phase
which is
always
continuing
but especially
during early
developmental

phases
uh
it's
catching on
to patterns
and rhythms
and whether
the the
four beat
measure
or different
pitches
and um
rhythms
are familiar
or not
that's
very
interesting
area
so
how would
one
model that
and or
where do you
see active
inference
in this
discussion
and what
you've
discussed
here
yeah
so there's
um
then predictive

coding approaches

at modeling

this

um

also the

information

dynamics

of music

um

it's one

model that

comes to

mind

um

or if you

think of

any of

these ai

music

generators

um

google has

an ai

that plays

music like

bach

um

actually that

might not

be google

but there

is an ai

that plays

music like

bach

um

google has

uh

different
music ais
that you
can actually
interact
with
and what
you do
here
is
you turn
all of
the music
into
symbols
for the
computer
um
and you're
going to
encode
what the
pitch is
you're going
to encode
what the
the rhythm
is
in a set
of however
many different
numbers you
want
and then
you can
kind of
take measures
of of

maybe the
shannon
entropy
of how
likely you
are to
expect
this note
knowing
that that
note
has
played
and if
you've
only
trained
this ai
on
bach
corrals
maybe
then you
know
the
notes
are
mostly
going to
be
quarter
notes
maybe
half
notes
maybe
an eighth
note

here
there
and
some
whole
notes
so you
already are
constrained
with the
type
the length
of note
that you're
going to hear
and back
corrals follow
a similar
a similar
pattern
every single
corral
so you
have
you know
the probability
of you
hearing
the tonic
chord
at this
point in
time
and not
hearing
a major
second
or

hearing
a
certain
step
at this
point in
time
because you've
mapped out
all those
probabilities
and lots
of people
do that
I don't
do that
kind
of
AI
or
modeling
work
with
musical
scores
myself
so
they're
creating
musical
scores
with
what
imperative
or
for
what
end

that's
a good
question
so
I've
actually
worked
with a
co-author
who's
submitted
a
manuscript
and that's
kind of
one of
the
questions
we ask
waiting
on reviews
back
but
we're
looking at
all of
these
different
kind of
music
making
Als
and
we're
coming at
the question
from another
angle

we're
kind of
asking
whether
they can
make
music
and whether
it's
good
music
and what's
missing
if they
can't
music
and
that's
because
when we're
talking
about
humans
and
why
humans
make
music
a lot
of
it
comes
down
to
the
social
interaction
that

music
allows
so
music
seems
to
facilitate
social
bonding
if
I
am
drumming
in
time
with
you
even
if
I'm
walking
in
time
with
you
but
if
I'm
doing
something
in
time
with
you
whether
it's
drumming
or

tapping
or
making
music
I
probably
will
like
you
more
after
that
event
and
it's
also a
tool
that we
use
parents
to
infants
to
soothe
them
if
you
think
about
lullabies
and
there's
a lot
about
music
that
seems
to

be
rooted
in
social
bonding
and
music
also
activates
reward
systems
in
our
brain
so
we
get
a
dopamine
rush
when
we
get
to
that
peak
experience
in
music
there's
some
recent
work
and
I
can't
remember
who

put
it
out
that
if
you
listen
to
music
you're
getting
similar
activation
as
if
you
had
engaged
in
exercise
we
use
music
in
therapy
like
music
is
just
it's
connected
to
a
lot
of
!
social
bonding

functions
for
humans
but
why
doesn't
AI
make
music
like
the
AI
doesn't
need
to
bond
with
another
AI
maybe
they're
getting
reward
in
terms
of
a
value
function
that
we've
programmed
into
the
AI
to
say
oh

good
that
was
a
good
musical
line
that
you
made
now
do
it
again
but
what
motivation
do
they
have
there's
also
no
connection
to
any
interoceptive
states
or
physiological
states
if we
think
about
listening
to
music
before

we
go
to
the
gym
we
listen
to
music
to
hype
us
up
and
get
us
pumped
and
it's
raising
our
blood
pressure
it's
getting
us
our
body
prepared
to
do
the
activity
we
want
to
do
and

at
night
we
might
listen
to
calming
soothing
music
lower
blood
pressure
lower
breathing
rate
to
prepare
to
go
to
sleep
what
drive
does
a
computer
have
to
your
life
compared
to
a
human
it's
a
great
question

I
don't
know
if
the
people
who
are
making
AI
music
are
always
asking
that
question
but
have
you
ever
heard
of
Holly
Herndon
no
so
she
is
a
electronic
music
creator
who
has
created
a
music
AI

in
this
space
and
her
notion
was
to
take
this
idea
of
cultural
learning
and
if
you
think
of
the
zone
of
proximal

development
but
she
trained
her
AI
baby
with
voices
of
a lot
of
different
people

and
exposed
to
a lot
of
like
basically
a
culture
in order
for it
to generate
some sort
of music
from what
she makes
and she
has some
really
interesting
art
in that
space
it's
a little
bit like
the tree
falling
in the
forest
is it
a noise
or a
sound
does
someone
have
to

listen
to
it
if
you
have
a
bunch
of
auto
generated
songs
in a
genre
but
no
human
experienced
writing
them
no one
experienced
listening
to them
it's like
did it
ever
happen
that's
quite an
interesting
idea
and it's
also
it's a
cross
media
issue

with
generated
text
and image
and
that also
makes me
think about
a John Cage
quote
talking about
music
and duration
and time
so maybe
what do
you
think
about
music
and time
music
and time
so
stereotypically
whenever we
think of
John Cage
think of
that
four minutes
and 33
seconds
he sits
in front
of the
piano
starts

a timer
and is
just
silent
there's
a
Marius Kozak
if I'm
remembering
right
actually
writes
about
music
and time
and the
idea
that
music
is
deeply
linked
to
movement
in time
and it's
sort of
a way
to
express
movement
in time
through
not just
the auditory
lens
but also
through

dance
so
like
thinking
of dance
as a
form
of music
as well
and seeing
how
not
only
something
like
auditory
music
but dance
is prevalent
in every
culture
in the
globe
and I
think
there's
something
important
to
the
notion
of
time
and
maybe
music
as
the

choreographing

of

movement

through

time

because

if we

think

about

you

mentioned

generated

language

or images

these

are

static

pieces

you know

you can

look

around

the scene

or you

can

read

the

text

but

music

has

to

happen

in the

temporal

domain

you

have

to
listen
to
it
over
a
period
of
time
once
the note
is
played
it's
gone
after
that
or
once
you've
enacted
a
particular
dance
it's
gone
after
that
and
I
don't
know
I
don't
have
like
a
nice

way
to
wrap
that
up
about
why
time
is
important
but
it's
definitely
an
important
feature
of
music
and
of
dance
that
you're
not
going
to
find
in
other
art
forms
you
might
find
you
could
think
of

it
in
a
narrative
has
to
happen
over
time
because
you're
reading
it
over
time
and
that's
part
of
why
so
in
my
neuroscience
work
I
study
how
we
perceive
musical
rhythm
so
if
we
perturb
the
time

that
a
musical
rhythm
is
happening
how
does
our
brain
cope
with
that
or
if
we
perturb
part
of
the
brain
that
is
thought
to
be
involved
with
timing
perception
so
this
ability
to
hear
beats
that
are

happening

in

a

certain

phase

or

to

hear

certain

rhythms

if

we

perturb

that

system

are

we

going

to

impair

how

people

perceive

the

rhythm

in

that

piece

of

music

and

these

are

questions

I'm

asking

questions

a lot

of
people
are
asking
it's
a
very
bright
landscape
right
now
for
that
kind
of
question
that's
really
interesting
about
bringing
the
action
in
and
maybe
even
moving
music
outside
or
a little
bit
beyond
just
the
auditory
because

then that
might open
up
auditory
experiences
that
aren't
music
and
musical
experiences
that have
a large
component
through time
but they're
not necessarily
auditory
so one
other
thing
thinking
about
the
development
of
auditory
preferences
and
expectations
for
probably
a long
time
in human
history
the
sounds

and the
music
that one
heard
was local
and
performed
if not
interactive
and then
there was
a pretty
short
period
of time
of
top
40
records
CDs
and then
now
there's
the
ability
to
listen
to
a
huge
number
of
music
as well
as
listen
to
a

small
selection
many
times
and
what
is that
doing
to
our
auditory
cognition
or
even
more
broadly
that
I can
listen
to
the
same
song
with
50
different
versions
in a
row
versus
that
just
not
being
plausible
not
even
so

long
ago
yeah
that's
an
interesting
question
I
haven't
thought
a lot
about
it
but
I'm
basically
distracting
from
music
in the
home
to
everybody
hearing
everything
the same
thing
on the
radio
to
now
we
have
Spotify
and
the
internet
and

it's
probably
a question
that's
not just
about
music
it's
also
about
media
we
went
from
only
getting
our
media
or
news
from
people
that
we
locally
interacted
with
everybody
sitting
for
the
same
fireside
chat
so
now
everybody
is

in
their
different
bubble
for
whatever
media
they're
consuming
and
it's
interesting
I
know
there's
been
some
work
for
instance
Mike
Hove
has
looked
at
how
the
bass
sounds
in
music
have
gotten
lower
over
time
lower
and

louder
which
has
some
impact
on
how
we
feel
musical
groove
since
the
bass
is
usually
where
the
rhythm
sits
in
pop
music
but
yeah
so
what
is it
doing
to
our
brains
and
our
perception
of
music
that's

interesting
and you
mentioned
memes
earlier
which
made me
think
of
TikTok
and
so
this
new
way
of
proliferating
sounds
and
if
you've
ever
spent
very
much
time
on
TikTok
there'll
be
like
one
sound
that
everybody's
making
a
dance

to
or
one
sound
that
people
are
just
playing
over
the
top
of
the
video
of
their
dogs
running
around
because
that
sounds
popular
and
then
when
you
hear
it
on
the
radio
it's
a
little
bit
jarring

because
you
only
know
this
like
15
second
snippet
or
however
long
a
TikTok
is
from
that
TikTok
video
and
then
you
realize
this
is
a
real
song
like
a
real
work
of
art
that
somebody
made
before

it
became
a
TikTok
meet
which
is
just
pure
personal
anecdote
but
it's
interesting
so
if
we're
getting
insulated
by
the
media
we
consume
are
we
also
getting
insulated
by
the
type
of
music
we
consume
like
are

we
less
able
to
appreciate
other
types
of
music
because
I'm
listening
to
the
same
song
in
a lot
of
different
renditions
or
are
we
broadening
our
musical
consumption
because
I'm
able
to
access
music
from
all
over
the

globe
at
the
click
of
a
button
it's
probably
a little
bit
of
both
yeah
it's
like
the
space
of
the
action
opens
up
when
there's
so many
affordances
in our
niche
to select
from
yet
also
quite
curated
and
presented
in a

certain
way
that
just
on
aggregate
probably
still
does
drive
people
to
have
shared
action
and
so
then
like
it's
literally
the
platform
playing
that
role
of
the
metronome
coupling
interaction
there
that's
interesting
that you
bring
that up
and

also
so
if we
come to
this
music
is
this
drive
for
social
bonding
and
we've
just
been
in
this
global
pandemic
or
we
still
are
but
for
many
people
we
were
on
various
stages
of
lockdowns
music
live
music

was
not
happening
choir
or
musical
ensemble
rehearsals
were
not
happening
and
you
saw
this
drive
to
continue
to
do
something
like
make
music
together
online
and
one
of
these
drives
was
she
shanties
became
really
popular
on

TikTok
and
it's
not
something
that
people
were
doing
together
at
the
same
time
because
we
have
this
delay
that's
happening
over
audiovisual
communication
like
this
it works
well
when
we're
taking
turns
it's
harder
for
trying
to
synchronize

and
sing
together
so
TikTok
allowed
someone
to
record
a
sea
shanty
and
then
the
next
person
adds
their
voice
to
that
recording
and
they
record
their
song
singing
the
sea
shanty
and
then
the
next
person
does

the
same
thing
and
you
end
up
with
these
compilations
not
compilations
but
these
ensemble
performances
of
a
song
that
people
would
sing
when
they're
off
on
ships
at
sea
and
they're
feeling
they're
having
this
social
bond

with
the
people
that
they're
out
at
sea
with
the
whole
time
and
now
it's
our
song
that
we're
singing
through
the
internet
as
a
way
to
feel
like
we're
having
some
sort
of
social
bonding
!
interacting

and
engaging
with
people
and
that
with
celebrities
trying
to sing
when they
all
gathered
together
to sing
Imagine
we can
argue
about
whether
that
went
well
or
not
or
when
we
saw
basically
there was
kind
of
explosion
of
different
types
of

online
music
interaction
to
try
to
replace
this
local
interaction
of going
to a
concert
and
seeing
someone
on
stage
around
a
group
of
people
that
you're
interacting
with
together
or
going
to
perform
or
jam
with
your
friends
in

your
garage
and
trying
to
find
digital
avenues
for
the
same
social
role
that
music
traditionally
plays
Very
interesting
like
and how
will it
change
going
forward
some
people
may
find
very
immersive
sensory
experiences
to
be
powerful
for
them

maybe
that's
a
scalable
approach
maybe
it's
not
maybe
that's
a
good
thing
maybe
it's
not
so
one
other
part
was
like
when
you
were
contrasting
the
musical
study
that
you
presented
first
with
the
basketball
you
mentioned

that
that
environment
was
unscripted
or
in
contrast
to
sheet
music
or
like a
spoken
dialogue
that
was
scripted
and
then
in
one
sense
it
is
unscripted
but
then
as
per
the
paper
on
active
inference
and
scripts
it's

more
like
the
strong
and
the
weak
scripts
like
the
basketball
is
a
scripted
environment
in
the
cultural
norm
weak
script
sense
so
yes
it doesn't
have
the
stage
directions
but
you
know
are
there
not
signs
helping
identify

where
people
with a
given
ticket
should
sit
and
signals
and
lights
and
sounds
that
also
provide
a lot
of
cues
that
are
going
to
be
specific
like
the
seventh
inning
stretch
in
baseball
why
stretch
then
but
then
that's

part
of
where
like
the
embodiment
of even
watching
the
game
and
participating
in it
gets
to be
why
it's
the
phenomena
of
entertainment
and
why
it's
something
that
people
spend
their
life
like
supporting
directly
and
why
this
isn't
just

like
a
neural
entrainment
binaural
beat
hack
or
something
like
that
yeah
so
if
I
got
the
same
experience
being
at
a
basketball
game
sitting
in
the
student
section
that
I
got
from
sitting
on
my
couch
and

watching
the
basketball
game
at
home
then
you
would
imagine
nobody
pays
to
go
to
basketball
games
and
we
all
sit
on
our
couch
and
watch
them
at
home
right
but
we
spend
lots
of
money
to
go

to
basketball
games
or
whatever
sporting
events
we
have
student
sections
we
wear
the
colors
of
our
sports
team
so
we're
already
signaling
to
people
that
you
know
I'm
wearing
this
outfit
so
I'm
a
BYU
fan
I'm

wearing
this
other
outfit
so
I'm
a
I
don't
know
basketball
teams
I was
going
to
name
another
one
I'm
that
kind
of
fan
so
that's
already
signifying
a
kind
of
script
where
I
know
oh
yeah
you're
a

friend
we're
going
to
cheer
at
the
same
time
and
then
the
game
has
rules
and
the
game
has
good
things
to
happen
if
my
team
scores
a
point
my
response
is
going
to
be
to
cheer
if

the
other
team
scores
a
point
I'm
going
to
be
upset
so
there's
already
a
relationship
where
the
game
is
and
in
front
of
you
the
people
who
you
can
hear
also
behind
you
and
there
are
certain

cheers
that
might
be
scripted
just
like
that
musical
ensemble
so
if
there's
a
certain
idiosyncratic
chant
that
I
make
that's
associated
with
this
team
and
this
mascot
we
all
know
this
chant
we've
learned
it
maybe
it

was
written
down
maybe
it's
being
broadcast
on
the
big
screen
above
the
game
maybe
there's
cheerleaders
or a
band
who's
leading
a
certain
cheer
and
you're
engaging
it
so
even
though
it's
less
scripted
in a
way
than
this

musical
ensemble
in the
sense
that
it's
not
written
on a
piece
of
paper
that
every
person
is
looking
at
they're
all
aware
of
the
social
scripts
that
are
involved
in
a
basketball
game
experience
and
a
basketball
game
experience

with
that
team
or
with
that
student
section
that
you're
sitting
in
so
you're
absolutely
right
that
there's
scripts
like
this
and
I
think
any
interaction
you
get
to
there's
going
to be
some
kind
of
script
there's
not

going
to
be
something
that's
pure
interaction
nobody's
interacting
just
on
vibes
you
know
there's

some
social
script
based
on
where
the
people
are
based
on
what
context
they're
in
whether
they
know
each
other
ahead
of

time
whether
there's
some
external
signal
like
clothing
or
you
look
like
me
or
you
don't
look
like
me
and
that's
all
going
to
change
the
dynamics
it
might
make
certain
forms
of
interactions
occur
more
easily
it

might
make
certain
kinds
of
interactions
dissipate
more
easily
which
interestingly
is
something
that
we're
still
studying
with
metronomes
so
metronomes
don't
only
move
into
synchrony
if
you
have
two
interacting
metronomes
they might
actually
cancel
each
other
out

or
they
might
exhibit
other
like
coming
in and
out
of
synchronous
dynamics
and
if
we
are
still
learning
how
to
understand
that
in
that
physical
system
where
there's
physical
properties
change
and
measure
that
are
really
simple
just

moving
back
and
forth
in a
certain
period
is that
much more
complicated
when we're
looking at
humans
who we
seem to
have some
sort of
free will
and we
can behave
according to
some
social rules
that we
know about
or we
could behave
completely
contrary
to those
rules
awesome
awesome
point
that the
model
system
like you

brought
up
helps
us
confirm
just
our
statistics
like
before
the
linear
model
gets
brought
out
in
one
setting
try it
on the
piece
of
graph
paper
and so
this
is
like
the
graph
paper
for
thinking
about
semantic
or
narrative

coordination
but
one
wouldn't
want to
be
in a
non
observable
territory
with
kind of
flying blind
that way
and also
I think
the
examples
with
the
game
and the
context
coordinating
behavior
to different
ends
it made me
think about
where there's
fans from
different
teams
in the
same
setting
versus
like

sections
that are
separated
and then
there's the
parents
at the
game
and maybe
like
the intention
is just
to support
people
in their
play
versus
like
professional
sports
and then
there's
like
people
who
are
there
for
pragmatic
value
they have
literal
money
other
people
have
like
identity

currency
on the
line
and
like
feeling
like
one's
city
has a
good
baseball
team
and all
these
very
complex
things
and so
it's
kind
of
just
cool
to
have
a
model
that
maybe
can
blanket
off
some
of
that
complexity
but

also
just
recognize
the
real
data
that
you're
getting
from
these
systems
which is
sometimes
as you
pointed
out
down
to
a
single
microphone
which
has
many
advantages
over
visual
observation
or like
sensor
tracking
or anything
and that's
what
we're
observing
from the

system
and then
there's
our
process
of
doing
analysis
on that
signal
because
that's
actually
what
the
system
is
providing
as
measurements
so
it
does
tie
in
active
inference
and
how
we're
modeling
these
complex
settings
without
getting
lost
in how

many
possible
ways
they
can
be
cool
do you
have any
other
things you'd
like to
add
or any
other
points
you want
to
bring
up
no
no
I think
this was
really
great
I
really
appreciated
the
discussion
you
brought
we
covered
a lot
more
than

just
interacting
groups
of
people
running
from
brains
to
music
Als
back
again
to
people
it
was
really
fun
cool
thank
you
very
much
for
joining
so
see
you
anytime
awesome
sounds
!
Good
see
you
bye
good

Thank you.